Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The straight lines ℓ_1, ℓ_2, ℓ_3 are parallel and lie in the same plane. A total number of m points are taken on ℓ_1, n points on ℓ_2, k points on ℓ_3. The maximum number of triangles formed with vertices at these points are :

(A) ${}^{m+n+k}\mathrm{C}_3$

(B) ${}^{m+n+k}\mathrm{C}_3 - {}^{m}\mathrm{C}_3 - {}^{n}\mathrm{C}_3 - {}^{k}\mathrm{C}_3$

(C) ${}^{m}\mathrm{C}_3 + {}^{m}\mathrm{C}_3 + {}^{k}\mathrm{C}_3$

(D) None of these

Q.2

The total number of 3 -digit numbers, the sum of whose digits is even, is equal to

(A) 450

(B) 350

(C) 250

(D) 325

Q.3

Two straight line intersect at a point O. Points $A_1, A_2, \ldots . A_n$ are taken on one line and points B_1, $B_2, \ldots ., B_n$ on the other. If the point O is not to be used, the number of triangles that can be drawn using these points as vertices, is

(A) $n(n-1)$

(B) $n(n-1)^2$

(C) $n^2(n-1)$

(D) $n^2(n-1)^2$

Q.4

Consider n points in a plane no three of which are collinear and the ratio of number of hexagon and octagon that can be formed from these n points is $4:13$ then find the value of n.

(A) 14

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(B) 20

(C) 28

(D) None of these

Q.5

In a plane there are 37 straight lines of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both points A and B, and no two are parallel. Then the number of intersection points the lines have is equal to

(A) 535

(B) 601

(C) 728

(D) None of these

Q.6

The sides AB, BC, CA of a trangle ABC, have 3,4 and 5 interior points respectively on them. The total number of triangles that can be constructed by using these points as vertices is

(A) 220

(B) 204

(C) 205

(D) 195

Q.7

The greatest possible number of points of intersection of 8 straight lines and 4 circles is

(A) 32

(B) 64

(C) 76

(D) 104

Q.8

The number of triangles whose vertices are at the vertices of an octagon but none of whose sides happen to be common with the sides of the octagon is

(A) 24

(B) 52

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(C) 48

(D) 16

Q.9

Let T_n denote the number of triangles which can be formed by using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 28$, then n equals

(A) 4

(B) 5

(C) 6

(D) 8

Q.10

Given five line segments of length $2, 3, 4, 5, 6$ units. Then the number of triangles that can be formed by joining these lines is

(A) ${}^5C_3 - 3$

(B) ${}^5C_3 - 1$

(C) ${}^5C_3$

(D) ${}^5C_3 - 2$

Q.11

There are three coplanar parallel lines. If any p points are taken on each of the lines, the maximum number of triangles with vertices at these points is

(A) $3p^2(p-1)+1$

(B) $3p^2(p-1)$

(C) $p^2(4p-3)$

(D) None of these

Q.12

The number of triangles whose vertices are at the vertices of an octagon but none of whose sides happen to come from the sides of the octagon is

(A) 24

(B) 52

(C) 48

(D) 16

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Q.13

A parallelogram is cut by two set of n parallel lines, parallel to the sides of the parallelogram. The number of parallelograms formed is:

(A) $\left({}^{n+2}C_2\right)\left({}^{n+2}C_2\right)$

(B) $\left({}^{n}C_2\right)\left({}^{n+1}C_2\right)$

(C) $\left({}^{n}C_2\right)\left({}^{n+2}C_2\right)$

(D) $\left({}^{n}C_2\right)\left({}^{n}C_2\right)$

Q.14

The maximum possible number of points of intersection of 8 straight lines and 4 circles is

(A) 164

(B) 76

(C) 104

(D) 32

Q.15

There are 3 set of parallel lines containing respectively p lines, q lines and r lines respectively. The greatest number of parallelograms that can be formed by the system

(A) $pqr+(p-1)(q-1)(r-1)$

(B) $\frac{1}{4}\{pqr+(p-1)(q-1)(r-1)\}$

(C) $\frac{1}{4}pqr(p+1)(q+1)(r+1)$

(D) none of these

Answers & Solutions

Q.1 Answer:

${}^{m+n+k}\mathrm{C}_3-{}^{m}\mathrm{C}_3-{}^{n}\mathrm{C}_3-{}^{k}\mathrm{C}_3$

Solution:

The straight line l_1, l_2, l_3 are parallel and lie in the same plane. Total number of points $= m+n+k$ Total no. of triangles formed with vertices $= {}^{m+n+k}\mathrm{C}_3$ By joining three given points on the same line we don't obtain a triangle. Therefore, the max. number of triangles $= {}^{m+n+k}\mathrm{C}_3-{}^{m}\mathrm{C}_3-{}^{n}\mathrm{C}_3-{}^{k}\mathrm{C}_3$

Q.2 Answer:

450

Solution:

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Let the number be $n = pqr$. As $p + q + r$ is even, there are following cases:
(i) p, q, r all are even $\rightarrow 4 \cdot 5 \cdot 5 = 100$ ways
(ii) p even and q, r are odd $\rightarrow 4 \cdot 5 \cdot 5 = 100$ ways
(iii) p odd, q and r even $\rightarrow 5 \cdot 5 \cdot 5 = 125$ ways
(iv) p odd, q even, r odd $\rightarrow 5 \cdot 5 \cdot 5 = 125$ Total ways $= 100 + 100 + 125 + 125 = 450.$

Q.3 Answer:

$n^2(n-1)$

Solution:

No. of triangles $= {}^{2n}C_3 - {}^{n}C_3 - {}^{n}C_3$
$= \frac{2n(2n-1)(2n-2)}{6} - \frac{2n(n-1)(n-2)}{6}$
$= \frac{1}{3}n(n-1)(3n) = n^2(n-1)$

Q.4 Answer:

20

Solution:

From n points number of hexagon is ${}^{n}C_6 = \frac{n!}{\{(n-6)!\}\{6!\}}$
From n points number of Octagons is
${}^{n}C_8 = \frac{n!}{\{(n-8)!\}\{8!\}}$
Ratio of number of hexagon to number of octagon is
$\frac{\{(\mathrm{n}-8)!\}\{8!\}}{\{(\mathrm{n}-6)!\}\{6!\}} = \frac{7\times 8}{(\mathrm{n}-7)(\mathrm{n}-6)} = \frac{4}{13}$
On solving this quadratic equation we will get $n = 20$

Q.5 Answer:

535

Solution:

The number of points of intersection of 37 straight lines is ${}^{37}\mathrm{C}_2$. But 13 of them pass through the point A. Therefore instead of getting ${}^{13}\mathrm{C}_2$ points we get merely one point. Similary 11 straight lines out of the given 37 straight lines intersect at B. Therefore instead of getting ${}^{11}\mathrm{C}_2$ points, we get only one point. Hence, the number of intersection points of the lines is

$$ {}^{37}\mathrm{C}_2 - {}^{13}\mathrm{C}_2 - {}^{11}\mathrm{C}_2 + 2 = 535 $$

Q.6 Answer:

205

Solution:

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We have in all 12 points.

Since 3 points form a triangle.

$\therefore$ the total number of triangles, including the triangles formed by collinear points on AB, BC and CA, is ${}^{12}C_3 = \frac{12\times11\times10}{1\times2\times3} = 220$

But this includes the number of triangles formed by 3 points on AB i.e. ${}^3C_3 = 1$ and for 4 points on $BC = {}^4C_3 = 4$ and 5 points on BC i.e. ${}^5C_3 = \frac{5\times4}{1\times2} = 10$

$\therefore$ Required number of triangles $= 220 - 1 - 4 - 10 = 205$

Q.7 Answer:

104

Solution:

The required number of points is ${}^8C_2 \times 1 + {}^4C_2 \times 2 + \left({}^8C_1 \times {}^4C_1\right) \times 2$

$$= 28 + 12 + 32 \times 2 = 104$$

Q.8 Answer:

16

Solution:

The number of triangles

= Total number of triangles

- Number of triangles having one side common with the octagon

- Number of triangles having two sides common with the octagon

$= {}^8C_3 - {}^8C_1 \times {}^4C_1 - 8 = 16$

Q.9 Answer:

8

Solution:

$\mathrm{T_n}$ denotes the number of triangles which can be formed by using the vertices of a regular polygon of n sides. $\therefore \mathrm{T_n} = {}^{\mathrm{n}}\mathrm{C}_3 \Rightarrow \mathrm{T_{n+1}} = {}^{\mathrm{n+1}}\mathrm{C}_3$

$\Rightarrow \mathrm{T_{n+1}} - \mathrm{T_n} = {}^{\mathrm{n+1}}\mathrm{C}_3 - {}^{\mathrm{n}}\mathrm{C}_3 = 28$ (Given)

$\frac{(n+1)!}{3!(n+1-3)!} - \frac{n!}{3!(n-3)!} = 28$

$\Rightarrow \frac{(\mathrm{n+1})\mathrm{n}(\mathrm{n-1})}{6} - \frac{\mathrm{n}(\mathrm{n-1})(\mathrm{n-2})}{6} = 28$

$\Rightarrow \frac{\mathrm{n}(\mathrm{n-1})}{6} \times 3 = 28 \Rightarrow \mathrm{n}(\mathrm{n}-1) = 28 \times 2$

$\Rightarrow (\mathrm{n}-8)(\mathrm{n}+7) = 0 \Rightarrow \mathrm{n} = 8, -7$

n can never be less than zero $\Rightarrow \mathrm{n} = 8$

Q.10 Answer:

${}^5\mathrm{C}_3 - 3$

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Solution:

We know that in any triangle the sum of two sides is always greater than the third side.

$\therefore$ The triangle will not be formed if we select segments of length $(2, 3, 5), (2, 3, 6)$ or $(2, 4, 6)$

Hence no. of triangles formed $= {}^{5}C_{3} - 3$

Q.11 Answer:

$p^2(4p - 3)$

Solution:

There are $3p$ points, so possible no. of triangles $= {}^{3p}C_{3}$

But the points on he same line do not form a triangle, such selections are $= 3.{}^{P}C_{3}$

$\therefore$ Desired number $= {}^{3p}C_{3} - 3.{}^{p}C_{3} = p^2(4p - 3)$

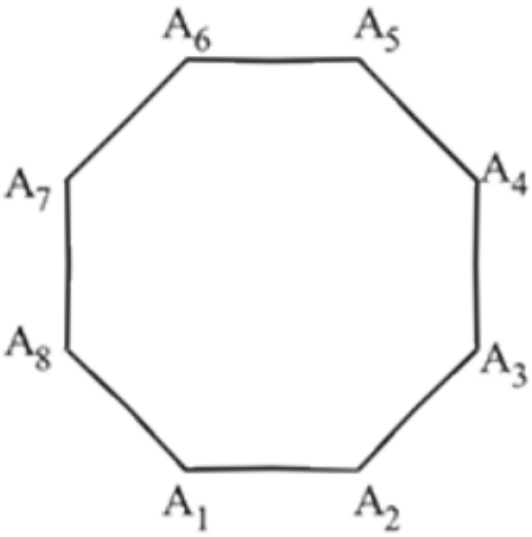
Q.12 Answer:

16

Solution:

Number of all possible triangles

= Number of selections of 3 points from 8 vertices $= {}^{8}C_{3} = 56$



Number of triangle with one side common with octagon $= 8 \times 4 = 32$

(Consider side A_1 A_2. Since two points A_3, A_8 are adjacent, 3rd point should be chosen from remaining points.) Number of triangles having two sides common with octagon : All such triangles have three consecutive vertices, viz., $A_1A_2A_3, A_2A_3A_4, \ldots..A_8A_1A_2$

Number of such triangles $= 8$

$\therefore$ Number of triangles with no side common $= 56 - 32 - 8 = 16.$

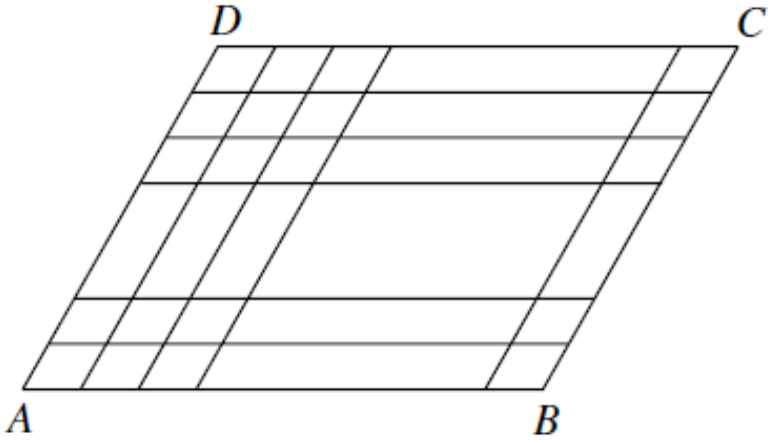
Q.13 Answer:

$\left({}^{n+2}C_{2}\right)\left({}^{n+2}C_{2}\right)$

Solution:

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There are $(n+2)$ line segments from A to B and $(n+2)$ line segments from A to D.



To get a parallelogram, choose two line segments from A to B and two line segments from A to D.

$\therefore$ number of required ways

$= \left({}^{n+2}C_2\right)\left({}^{n+2}C_2\right)$

Q.14 Answer:

104

Solution:

Number of points of intersections

$= \left({}^{8}C_2\right) + 2\left({}^{4}C_2\right) + 2\left({}^{8}C_1\right)\left({}^{4}C_1\right)$

$= 104$

Q.15 Answer:

none of these

Solution:

Number of parallelograms

$= \left({}^{p}C_2\right)\left({}^{q}C_2\right) + \left({}^{q}C_2\right)\left({}^{r}C_2\right) + \left({}^{r}C_2\right)\left({}^{p}C_2\right)$

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